In-Class Quiz: May 8

Points: 8/8

✓ **Correct** 1/1 Points

1. According to the mathematical model of an ideal pipeline, how does the optimal number of pipeline stages (k) relate to the latch delay (l)?

- ○ It is directly proportional to "l"
- ◉ It is inversely proportional to the square root of "l"
- ○ It is directly proportional to l^2
- ○ It is independent of "l"

✓ **Correct** 1/1 Points

2. As the amount of algorithmic work (t{max}) increases, what happens to the optimal number of pipeline stages for ideal performance?

- ○ It decreases
- ◉ It increases
- ○ It remains constant
- ○ It drops to 1

✓ **Correct** 1/1 Points

3. For an ideal pipeline with "n" instructions and "k" stages, what is the formula for the Cycles Per Instruction (CPI)?

- (n+k-1) / n
- (n+k) / k
- n / (n+k-1)
- (n-1+k) / k

✓ **Correct** 1/1 Points

4. In an ideal pipeline model, as the number of instructions (n) tends to infinity, what happens to the ideal number of pipeline stages?

- It tends to 1
- It tends to a constant value
- It tends to infinity
- It tends to t{max}

✓ **Correct** 1/1 Points

5. In reality, pipelines have stalls. How is the actual CPI of a non-ideal pipeline calculated?

- CPI{ideal} - (stall_rate X stall_penalty)
- CPI{ideal} X stall_rate
- CPI{ideal} / stall_penalty
- CPI{ideal} + (stall_rate X stall_penalty)

✓ **Correct** 1/1 Points

6. In the mathematical model presented for a non-ideal pipeline, the stall penalty is assumed to be proportional to which of the following?

- The number of instructions (n)

- [x] The number of pipeline stages (k)
- [ ] The latch delay (l)
- [ ] The frequency (f)

✓ **Correct** 1/1 Points

7. According to the non-ideal pipeline model, how should the optimal number of pipeline stages be adjusted for a program with a high number of dependences (a high stall rate, 'r')?

- [ ] Increase the number of stages
- [x] Decrease the number of stages
- [ ] Keep the number of stages constant
- [ ] Make the number of stages equal to the number of instructions

✓ **Correct** 1/1 Points

8. Compared to a pipeline that just has interlocks, how does adding data forwarding affect the optimal number of pipeline stages?

- [ ] It requires a smaller number of pipeline stages.
- [x] It requires a larger number of pipeline stages.
- [ ] It does not change the optimal number of pipeline stages.
- [ ] It eliminates the need for pipeline stages altogether.